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ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

DEPARTMENT OF MATHEMATICS

MATHEMATICS FOR NATURAL SCIENCES (MATH 1007)

Test-II

Date: July, 2022 Time allowed: 50 min Maximum Score 15 %

Name Abel Tessema **Id No.** ETS0027/14 **Section** A1

Part I: Short Answer Questions (10 points)
Write the Most simplified answer in the space provided [1.5 pts each]

1. If $f(x) = x^2 - 3x$ and $g(x) = 1 - x$, then $f(g(x)) =$ $x^2 - 5x + 1$
2. **The reminder of $P(x) = 5x^3 - 10x^2 + 9x - 2$ divided by $-1 + x$ is** 2.
3. **If $logx = 0.2$, and $logy = 0.3$, then $log\frac{x^3y}{\sqrt{xy}} =$** 0.65.
4. **The range of $f(x) = \frac{2x+1}{1-3x}$ is** $\mathbb{R} \setminus \{-\frac{2}{3}\}$.
5. The simplified form of $f(x) = 9^{log_3^x}$ is $f(x) = x^2$.
6. The real and imaginary part of $z = 1 - \frac{1-i}{3i}$ are $\frac{4}{3}$ and $\frac{1}{3}$ respectively.

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Part II: Workout Questions [6 pts]

1. Find the cube roots of the complex number $z = 2 - 2i$
2. Prove that $n^3 + 2n$ is divisible by 3 for every natural number n.(use Mathematical induction)
3. For the function $f(x) = 2^{x+1}$,

   Find

   i) Domain

   ii) Range

   iii) Intercepts

   iv) Sketch the graph

<u>Answers for Part II</u>

1) $z = 2 - 2i$

$\theta = \tan^{-1}\left(\frac{-2}{2}\right) = \tan^{-1}(-1) = -\pi/4$

$r = \sqrt{2^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$

$z = \sqrt{8}\,[\cos(-\pi/4) + i\sin(-\pi/4)]$

$z^{1/3} = (8^{1/2})^{1/3}\left[\cos\left(\frac{1}{3}\times\left(\frac{-\pi}{4}\right)\right) + i\sin\left(\frac{1}{3}\times\left(\frac{-\pi}{4}\right)\right)\right]$

$= \sqrt[6]{8}\,[\cos(-\pi/12) + i\sin(-\pi/12)]$

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$z = \sqrt{8}\,e^{-\pi/4\, i}$

$z^{1/3} = (\sqrt{8})^{1/3}\, e^{1/3\times\left(\frac{-\pi i}{4}\right)} = \sqrt[6]{8}\, e^{-\frac{\pi i}{12}}$

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3) $f(x) = 2^{x+1}$

i) Domain $= \mathbb{R}$

ii) Range $= \mathbb{R}^{+} = (0, \infty)$

iii) $x_{int} \Rightarrow 0 = 2^{x+1} \Rightarrow \nexists$

$y_{int} \Rightarrow y = 2^{0+1} = 2 \Rightarrow (0, 2)$

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iv)